

3.2 Montel's theorem

Let Ω be an open subset of $\mathbb{C}$. A family $\mathcal{F}$ of holomorphic functions on Ω is said to be **normal** if every sequence in $\mathcal{F}$ has a subsequence that converges uniformly on every compact subset of Ω (the limit need not be in $\mathcal{F}$).

The proof that a family of functions is normal is, in practice, the consequence of two related properties, uniform boundedness and equicontinuity. These we shall now define.

The family $\mathcal{F}$ is said to be **uniformly bounded on compact subsets of Ω** if for each compact set $K \subset \Omega$ there exists $B > 0$, such that

$$|f(z)| \leq B \quad \text{for all } z \in K \text{ and } f \in \mathcal{F}.$$

Also, the family $\mathcal{F}$ is **equicontinuous** on a compact set K if for every $\epsilon > 0$ there exists $\delta > 0$ such that whenever $z, w \in K$ and $|z - w| < \delta$, then

$$|f(z) - f(w)| < \epsilon \quad \text{for all } f \in \mathcal{F}.$$

Equicontinuity is a strong condition, which requires uniform continuity, uniformly in the family. For instance, any family of differentiable functions on $[0, 1]$ whose derivatives are uniformly bounded is equicontinuous. This follows directly from the mean value theorem. On the other hand, note that the family $\{f_n\}$ on $[0, 1]$ given by $f_n(x) = x^n$ is *not* equicontinuous since for any fixed $0 < x_0 < 1$ we have $|f_n(1) - f_n(x_0)| \rightarrow 1$ as n tends to infinity.

The theorem that follows puts together these new concepts and is an important ingredient in the proof of the Riemann mapping theorem.

Theorem 3.3 *Suppose $\mathcal{F}$ is a family of holomorphic functions on Ω that is uniformly bounded on compact subsets of Ω . Then:*

- (i) $\mathcal{F}$ is equicontinuous on every compact subset of Ω .
- (ii) $\mathcal{F}$ is a normal family.

The theorem really consists of two separate parts. The first part says that $\mathcal{F}$ is equicontinuous under the assumption that $\mathcal{F}$ is a family of *holomorphic* functions that is uniformly bounded on compact subsets of Ω . The proof follows from an application of the Cauchy integral formula and hence relies on the fact that $\mathcal{F}$ consists of holomorphic functions. This conclusion is in sharp contrast with the real situation as illustrated by the family of functions given by $f_n(x) = \sin(nx)$ on $(0, 1)$, which is

uniformly bounded. However, this family is not equicontinuous and has no convergent subsequence on any compact subinterval of $(0, 1)$.

The second part of the theorem is not complex-analytic in nature. Indeed, the fact that $\mathcal{F}$ is a normal family follows from assuming only that $\mathcal{F}$ is uniformly bounded and equicontinuous on compact subsets of Ω . This result is sometimes known as the Arzela-Ascoli theorem and its proof consists primarily of a diagonalization argument.

We are required to prove convergence on arbitrary compact subsets of Ω , therefore it is useful to introduce the following notion. A sequence $\{K_\ell\}_{\ell=1}^\infty$ of compact subsets of Ω is called an **exhaustion** if

- (a) K_ℓ is contained in the interior of $K_{\ell+1}$ for all $\ell = 1, 2, \dots$
- (b) Any compact set $K \subset \Omega$ is contained in K_ℓ for some ℓ . In particular

$$\Omega = \bigcup_{\ell=1}^{\infty} K_\ell.$$

Lemma 3.4 *Any open set Ω in the complex plane has an exhaustion.*

Proof. If Ω is bounded, we let K_ℓ denote the set of all points in Ω at distance $\geq 1/\ell$ from the boundary of Ω . If Ω is not bounded, let K_ℓ denote the same set as above except that we also require $|z| \leq \ell$ for all $z \in K_\ell$.

We may now begin the proof of Montel's theorem. Let K be a compact subset of Ω and choose $r > 0$ so small that $D_{3r}(z)$ is contained in Ω for all $z \in K$. It suffices to choose r so that $3r$ is less than the distance from K to the boundary of Ω . Let $z, w \in K$ with $|z - w| < r$, and let γ denote the boundary circle of the disc $D_{2r}(w)$. Then, by Cauchy's integral formula, we have

$$f(z) - f(w) = \frac{1}{2\pi i} \int_{\gamma} f(\zeta) \left[\frac{1}{\zeta - z} - \frac{1}{\zeta - w} \right] d\zeta.$$

Observe that

$$\left| \frac{1}{\zeta - z} - \frac{1}{\zeta - w} \right| = \frac{|z - w|}{|\zeta - z| |\zeta - w|} \leq \frac{|z - w|}{r^2}$$

since $\zeta \in \gamma$ and $|z - w| < r$. Therefore

$$|f(z) - f(w)| \leq \frac{1}{2\pi} \frac{2\pi r}{r^2} B |z - w|,$$

where B denotes the uniform bound for the family $\mathcal{F}$ in the compact set consisting of all points in Ω at a distance $\leq 2r$ from K . Therefore $|f(z) - f(w)| < C|z - w|$, and this estimate is true for all $z, w \in K$ with $|z - w| < r$ and $f \in \mathcal{F}$; thus this family is equicontinuous, as was to be shown.

To prove the second part of the theorem, we argue as follows. Let $\{f_n\}_{n=1}^\infty$ be a sequence in $\mathcal{F}$ and K a compact subset of Ω . Choose a sequence of points $\{w_j\}_{j=1}^\infty$ that is dense in Ω . Since $\{f_n\}$ is uniformly bounded, there exists a subsequence $\{f_{n,1}\} = \{f_{1,1}, f_{2,1}, f_{3,1}, \dots\}$ of $\{f_n\}$ such that $f_{n,1}(w_1)$ converges.

From $\{f_{n,1}\}$ we can extract a subsequence $\{f_{n,2}\} = \{f_{1,2}, f_{2,2}, f_{3,2}, \dots\}$ so that $f_{n,2}(w_2)$ converges. We may continue this process, and extract a subsequence $\{f_{n,j}\}$ of $\{f_{n,j-1}\}$ such that $f_{n,j}(w_j)$ converges.

Finally, let $g_n = f_{n,n}$ and consider the diagonal subsequence $\{g_n\}$. By construction, $g_n(w_j)$ converges for each j , and we claim that equicontinuity implies that g_n converges uniformly on K . Given $\epsilon > 0$, choose δ as in the definition of equicontinuity, and note that for some J , the set K is contained in the union of the discs $D_\delta(w_1), \dots, D_\delta(w_J)$. Pick N so large that if $n, m > N$, then

$$|g_m(w_j) - g_n(w_j)| < \epsilon \quad \text{for all } j = 1, \dots, J.$$

So if $z \in K$, then $z \in D_\delta(w_j)$ for some $1 \leq j \leq J$. Therefore,

$$\begin{aligned} |g_n(z) - g_m(z)| &\leq |g_n(z) - g_n(w_j)| + |g_n(w_j) - g_m(w_j)| + \\ &\quad + |g_m(w_j) - g_m(z)| < 3\epsilon \end{aligned}$$

whenever $n, m > N$. Hence $\{g_n\}$ converges uniformly on K .

Finally, we need one more diagonalization argument to obtain a subsequence that converges uniformly on *every* compact subset of Ω . Let $K_1 \subset K_2 \subset \dots \subset K_\ell \subset \dots$ be an exhaustion of Ω , and suppose $\{g_{n,1}\}$ is a subsequence of the original sequence $\{f_n\}$ that converges uniformly on K_1 . Extract from $\{g_{n,1}\}$ a subsequence $\{g_{n,2}\}$ that converges uniformly on K_2 , and so on. Then, $\{g_{n,n}\}$ is a subsequence of $\{f_n\}$ that converges uniformly on every K_ℓ and since the K_ℓ exhaust Ω , the sequence $\{g_{n,n}\}$ converges uniformly on any compact subset of Ω , as was to be shown.

We need one further result before we can give the proof of the Riemann mapping theorem.

Proposition 3.5 *If Ω is a connected open subset of $\mathbb{C}$ and $\{f_n\}$ a sequence of injective holomorphic functions on Ω that converges uniformly*

on every compact subset of Ω to a holomorphic function f , then f is either injective or constant.

Proof. We argue by contradiction and suppose that f is not injective, so there exist distinct complex numbers z_1 and z_2 in Ω such that $f(z_1) = f(z_2)$. Define a new sequence by $g_n(z) = f_n(z) - f_n(z_1)$, so that g_n has no other zero besides z_1 , and the sequence $\{g_n\}$ converges uniformly on compact subsets of Ω to $g(z) = f(z) - f(z_1)$. If g is not identically zero, then z_2 is an isolated zero for g (because Ω is connected); therefore

$$1 = \frac{1}{2\pi i} \int_{\gamma} \frac{g'(\zeta)}{g(\zeta)} d\zeta,$$

where γ is a small circle centered at z_2 chosen so that g does not vanish on γ or at any point of its interior besides z_2 . Therefore, $1/g_n$ converges uniformly to $1/g$ on γ , and since $g'_n \rightarrow g'$ uniformly on γ we have

$$\frac{1}{2\pi i} \int_{\gamma} \frac{g'_n(\zeta)}{g_n(\zeta)} d\zeta \rightarrow \frac{1}{2\pi i} \int_{\gamma} \frac{g'(\zeta)}{g(\zeta)} d\zeta.$$

But this is a contradiction since g_n has no zeros inside γ , and hence

$$\frac{1}{2\pi i} \int_{\gamma} \frac{g'_n(\zeta)}{g_n(\zeta)} d\zeta = 0 \quad \text{for all } n.$$

3.3 Proof of the Riemann mapping theorem

Once we have established the technical results above, the rest of the proof of the Riemann mapping theorem is very elegant. It consists of three steps, which we isolate.

Step 1. Suppose that Ω is a simply connected proper open subset of $\mathbb{C}$. We claim that Ω is conformally equivalent to an open subset of the unit disc that contains the origin. Indeed, choose a complex number α that does not belong to Ω , (recall that Ω is proper), and observe that $z - \alpha$ never vanishes on the simply connected set Ω . Therefore, we can define a holomorphic function

$$f(z) = \log(z - \alpha)$$

with the desired properties of the logarithm. As a consequence one has, $e^{f(z)} = z - \alpha$, which proves in particular that f is injective. Pick a point $w \in \Omega$, and observe that

$$f(z) \neq f(w) + 2\pi i \quad \text{for all } z \in \Omega$$